Name- Heenal Dhanwani
Roll No- 2401024
Batch- T11
Subject- ASD&D
Branch- AI&DS

Aim:

To containerize application using Docker

LO 04:

To automate provisioning, configuration, container orchestration, and cloud resource management using Ansible, Docker, Kubernetes, and Terraform.

Theory:

What is Docker?

Docker is an open-source platform used to develop, package, ship, and run applications in containers. It allows developers to bundle an application along with all its dependencies, libraries, and configuration files into a single package, ensuring that the application runs consistently across different environments. Docker simplifies application deployment by eliminating compatibility issues between development, testing, and production systems.

What is Containerization?

Containerization is a virtualization technique in which an application and all its required dependencies are packaged into an isolated unit called a container. Containers share the host operating system's kernel but run independently from one another, making them lightweight and efficient compared to traditional virtual machines. Containerization ensures that applications behave the same regardless of the underlying operating system or infrastructure.

Difference between Image and Container

Docker Image

- Read-only template.
- Contains application and all required dependencies.
- Used to create containers.
- Static in nature.

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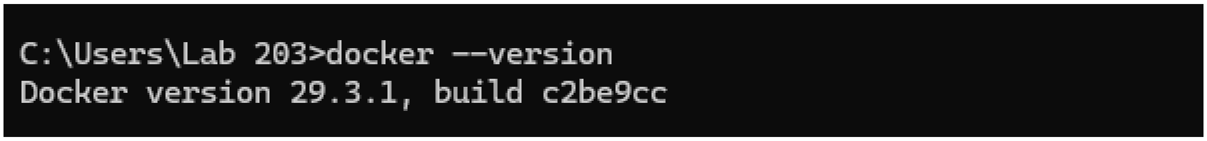
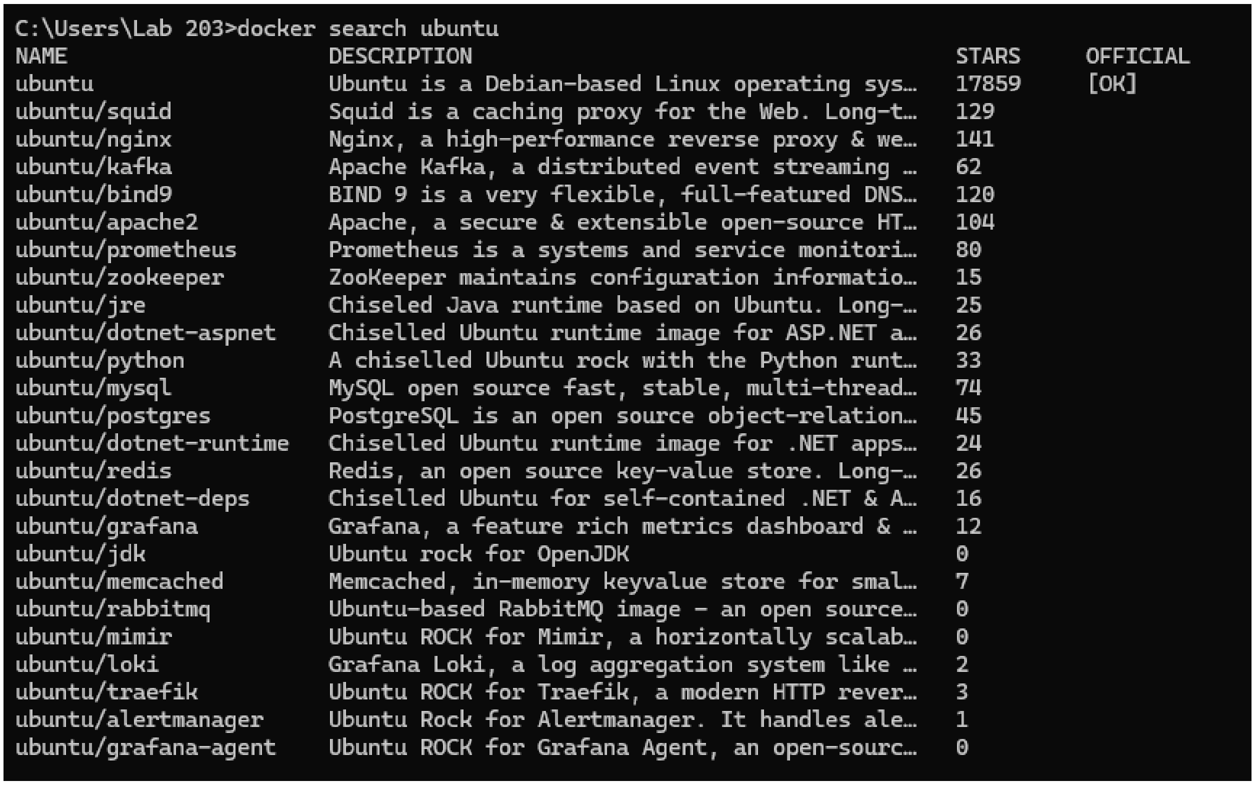
Docker Container

- Running instance of a Docker image.
- Executes the application.
- Dynamic and can change during execution.
- Consumes CPU, memory, and storage resources.

Commands:



```
C:\Users\Lab 203>docker --version
Docker version 29.3.1, build c2be9cc
```



```
C:\Users\Lab 203>docker search ubuntu
NAME                     DESCRIPTION                                     STARS     OFFICIAL
ubuntu                   Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid             Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx             Nginx, a high-performance reverse proxy & we…   141
ubuntu/kafka             Apache Kafka, a distributed event streaming …   62
ubuntu/bind9             BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/apache2           Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus        Prometheus is a systems and service monitori…   80
ubuntu/zookeeper         ZooKeeper maintains configuration informatio…   15
ubuntu/jre               Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet     Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python            A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql             MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres          PostgreSQL is an open source object-relation…   45
ubuntu/dotnet-runtime    Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/redis             Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-deps       Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana           Grafana, a feature rich metrics dashboard & …   12
ubuntu/jdk               Ubuntu rock for OpenJDK                         0
ubuntu/memcached         Memcached, in-memory keyvalue store for smal…   7
ubuntu/rabbitmq          Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir             Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki              Grafana Loki, a log aggregation system like …   2
ubuntu/traefik           Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/alertmanager      Ubuntu Rock for Alertmanager. It handles ale…   1
ubuntu/grafana-agent     Ubuntu ROCK for Grafana Agent, an open-sourc…   0
```

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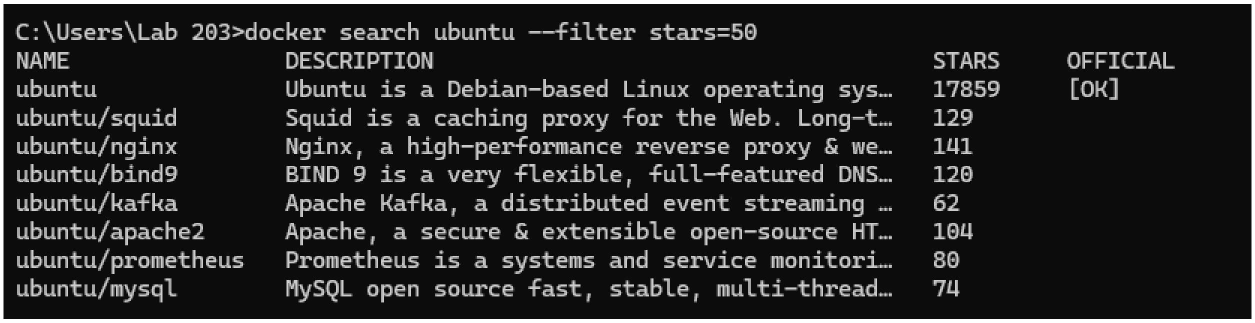
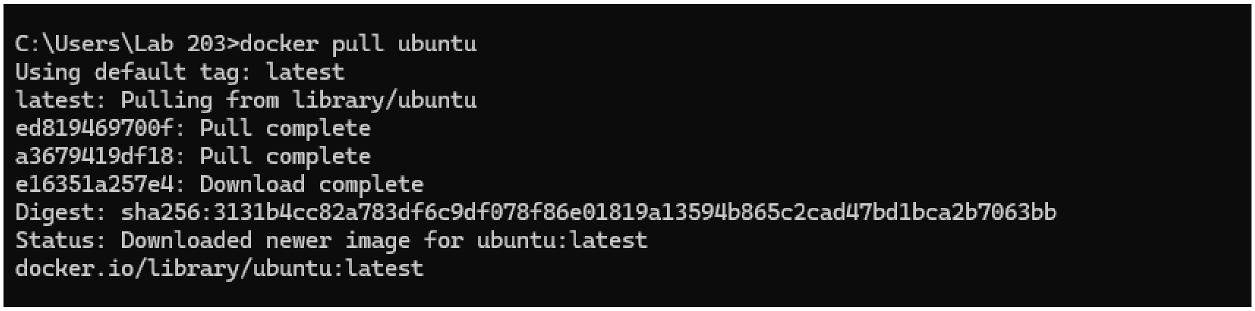
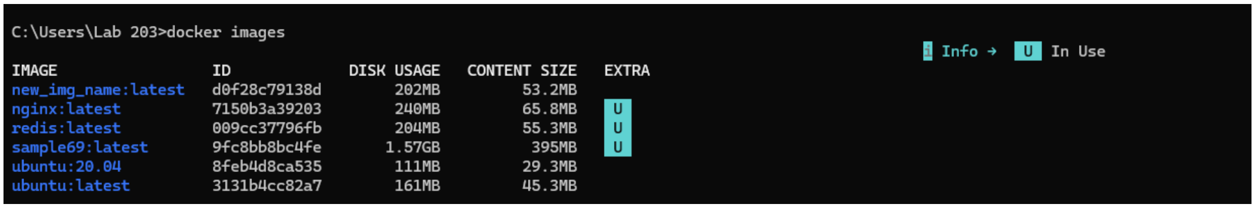


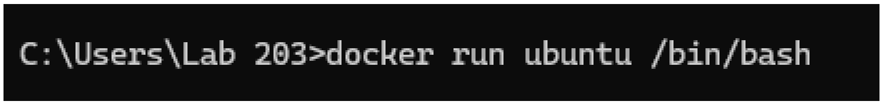
```
C:\Users\Lab 203>docker search ubuntu --filter stars=50
NAME                 DESCRIPTION                                     STARS     OFFICIAL
ubuntu               Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid         Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx         Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9         BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka         Apache Kafka, a distributed event streaming …   62
ubuntu/apache2       Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus    Prometheus is a systems and service monitori…   80
ubuntu/mysql         MySQL open source fast, stable, multi-thread…   74
```



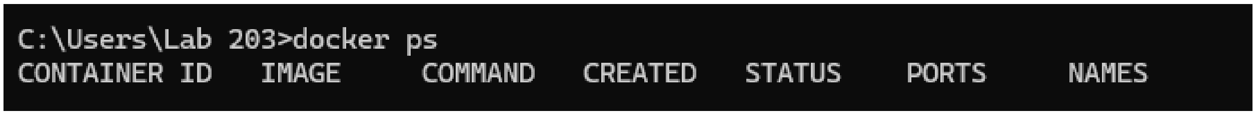
```
C:\Users\Lab 203>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
ed819469700f: Pull complete
a3679419df18: Pull complete
e16351a257e4: Download complete
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
```



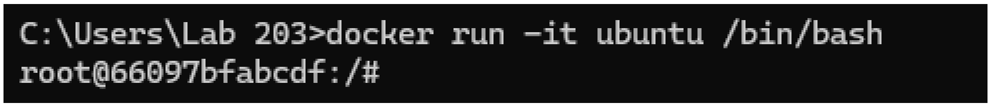
```
C:\Users\Lab 203>docker images
                                                                         i Info →   U In Use
IMAGE                 ID             DISK USAGE   CONTENT SIZE   EXTRA
new_img_name:latest   d0f28c79138d        202MB         53.2MB
nginx:latest          7150b3a39203        240MB         65.8MB    U
redis:latest          009cc37796fb        204MB         55.3MB    U
sample69:latest       9fc8bb8bc4fe       1.57GB          395MB    U
ubuntu:20.04          8feb4d8ca535        111MB         29.3MB
ubuntu:latest         3131b4cc82a7        161MB         45.3MB
```



```
C:\Users\Lab 203>docker run ubuntu /bin/bash
```



```
C:\Users\Lab 203>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```



```
C:\Users\Lab 203>docker run -it ubuntu /bin/bash
root@66097bfabcdf:/#
```

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Conclusion:

Docker containerization provides a standardized and efficient way to package and deploy applications. By including the application and all its dependencies in a single container, Docker eliminates compatibility issues and ensures consistent performance across different environments. Its lightweight architecture, portability, scalability, and ease of deployment make it an essential tool in modern software development and DevOps workflows.